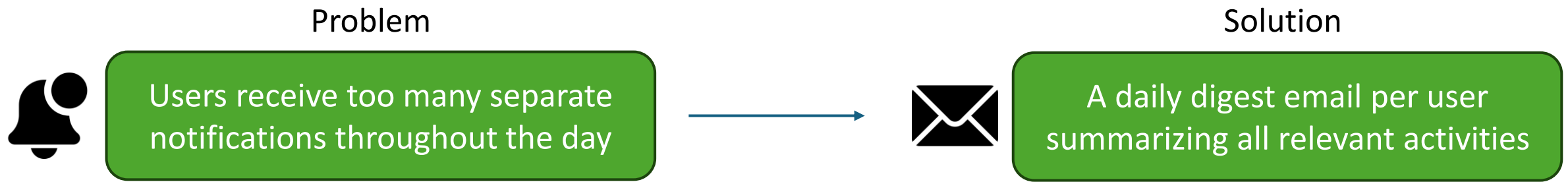


Design & Develop a solution of Daily Digest Email Notification in ServiceNow

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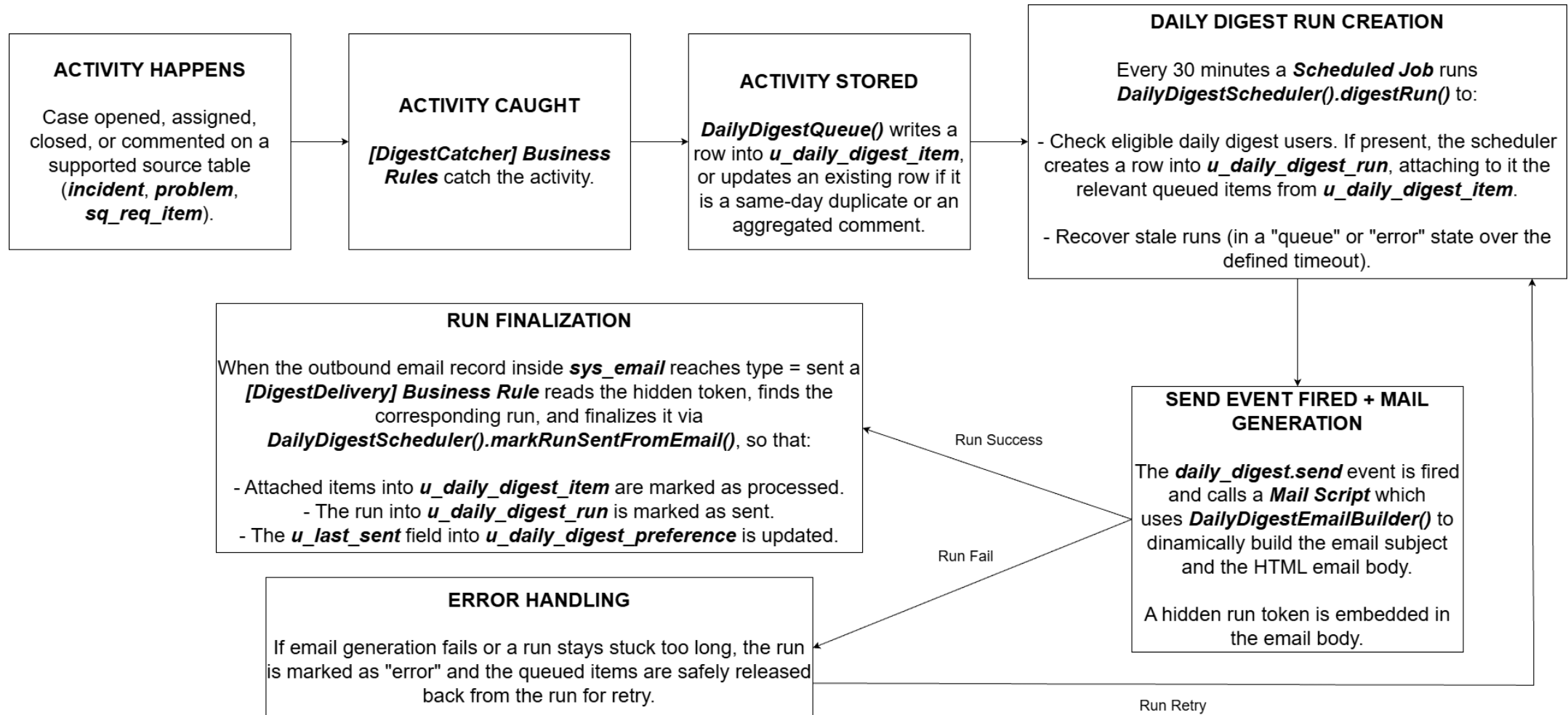
Rather than querying every source table at the end of the business day, I come up with a **3 layers** “*queue now and send later*” solution:

- Capture Layer Business Rules catch relevant events when they happen and queue them as digest items.
- Orchestration Layer Scheduled job decides who is eligible to receive a daily digest mail, consequently creating daily digest runs.
- Delivery Layer Notification and Mail Script build the daily digest email dynamically, and a finalization step marks it as successfully sent. The delivery happens at the end of business day in the recipient’s time zone.

Main Components

<i>Name</i>	<i>Type</i>	<i>Description</i>
u_daily_digest_preference	Table	Stores per user daily digest settings.
u_daily_digest_item	Table	Stores queued daily digest items.
u_daily_digest_run	Table	Stores daily digest runs (run = group of items).
DailyDigestConfig	Script Include	Reads system properties and provides shared config value.
DailyDigestQueue	Script Include	“Catches” activities via Business Rules into u_daily_digest_item.
DailyDigestScheduler	Script Include	Finds candidate users, creates and links items to runs, recovers runs.
DailyDigestEmailBuilder	Script Include	Takes one digest run and builds the corresponding final email.
[DigestCatcher]	Business Rule	Capture activities and turn them into queued daily digest items.
[DigestDelivery]	Business Rule	Detects sent emails and marks the related run items as processed.
Daily Digest Sched. Runner	Scheduled Job	Calls periodically the DailyDigestScheduler.
daily_digest.send	Event	Fires the Daily Digest Email Notification.
Daily Digest Email Notificat.	Email Notificat.	Turns the run event into the actual outgoing email.
daily_digest_body	Email Script	Builds the actual content of the daily digest email.

High Level Architecture



Capture Layer Details



Collect relevant daily activities and prepare them for the digest.

- *[DigestCatcher]* Business Rules **capture events**, when they happen during the day, on supported tables. Activities are stored in *u_daily_digest_item* and are **divided by type**: opened, assigned, closed, comments, approvals.
- Thanks to an unique key: **duplicates items are prevented** and **comments on the same case are aggregated** in the same *u_daily_digest_item* row.
- Daily digest items are later grouped considering the **user's local time zone**.

Orchestration Layer Details



Decide who is eligible to receive a digest, when, and with which content.

- Eligible users are found based on the following criteria: **presence of queued digest items**, **enabled preference** and **local send time**.
- Digest run creation is **per user** and **per local time window**. In addition, queued items in *u_daily_digest_item* are attached to a run in *u_daily_digest_run* thanks to a reference field: *u_digest_run*.
- Features a **recovery logic** for failed or stale runs, which calls *recoverStaleQueuedRuns()* function inside *DailyDigestScheduler* Script Include.

Delivery Layer Details



Generate the final daily digest email and confirm successful delivery.

- The **digest is sent through an event-based notification** (*daily_digest.send*) fired by the previous layer.
- A Mail Script is responsible of **building the final email content** calling the *buildMail()* function inside the *DailyDigestEmailBuilder Script Include*.
- The **email groups content into clear sections**: case opened by the recipient, case assigned to the recipient, case commented where the recipient is involved, case closed where the recipient were involved, pending approvals assigned to the recipient. A “show more...” link will appear in case of many records per section.
- An hidden run token in the email is needed for **finalize a run only when outbound email is actually marked as sent**, calling *finalizeRun()* inside *DailyDigestScheduler Script Include*.

Live Demo.

Thank you for your time!